# Tighter Bounds on the Expected Running Time of Ungarian Markov Chains

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# Lattices

## Definition

A *lattice* $L$ is a poset such that for any $x, y \in L$,

- $x, y$ have a greatest lower bound, $x \wedge y$, called the "meet".
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**Definition**

Given a lattice $L$ and $x, y \in L$, say $x$ *covers* $y$ ($x \gtrdot y$) if $y < x$ and $\nexists\, z \in L$ satisfying $y < z < x$.

**Definition**

$$\mathrm{cov}_L(x) = \{y : x \gtrdot y\}.$$

# Examples of Lattices

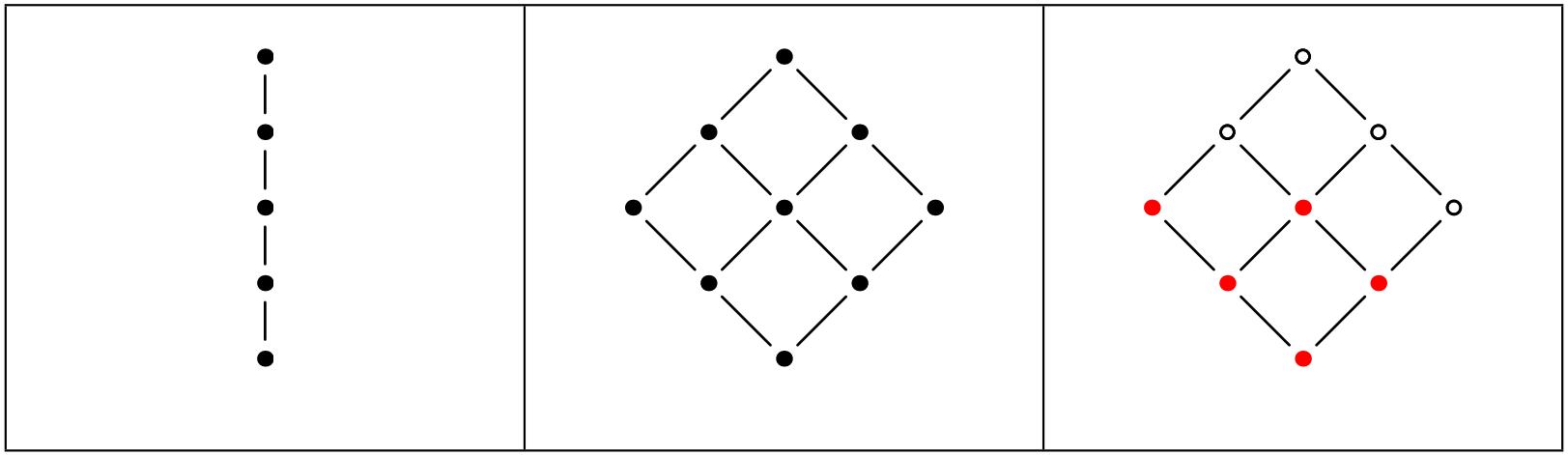


Figure: On the left is a chain. In the middle is a product of chains. On the right is a subset (in red) of the poset closed under the partial order; these subsets form a lattice under inclusion.

# Ungarian Markov Chains

Fix $p \in (0, 1)$.

### Definition (Defant–Li '23)

Given $x \in L$, select any element of $\mathrm{cov}_L(x)$ independently with probability $p$. Let $T \subset \mathrm{cov}_L(x)$ be the set of selected elements. Then a *random Ungar move* at $x$ sends

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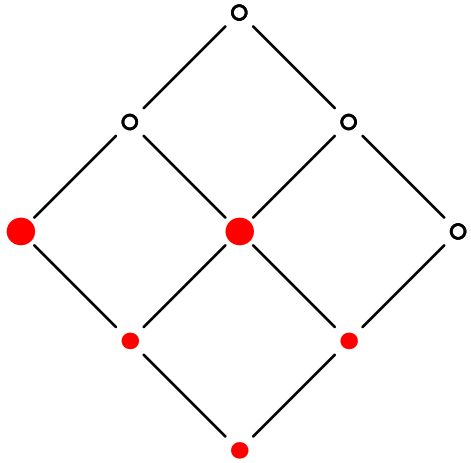
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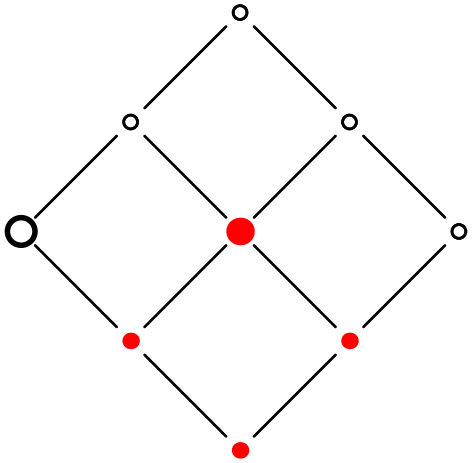
Given a lattice $L$, the *Ungarian Markov chain* $U_L$ on $L$ is the Markov chain with:

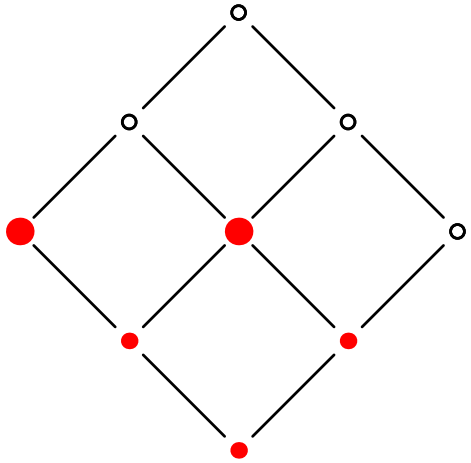
- state space $L$, and
- transitions given by the random Ungar moves.

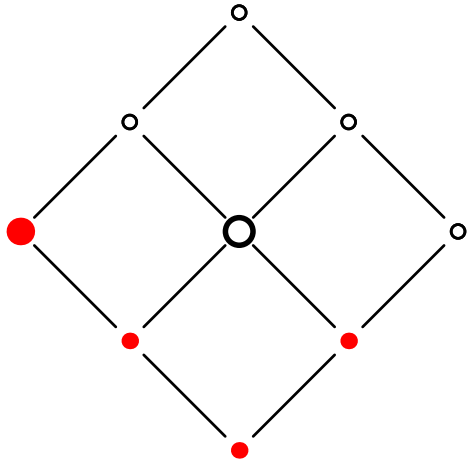
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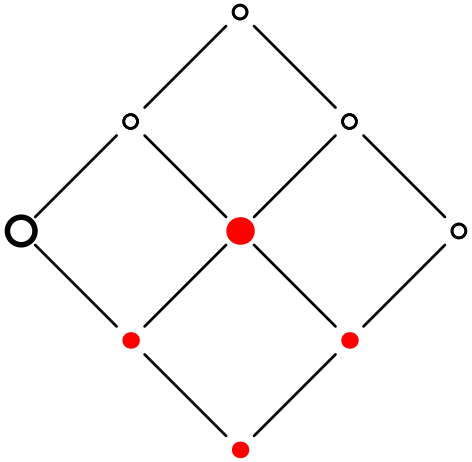
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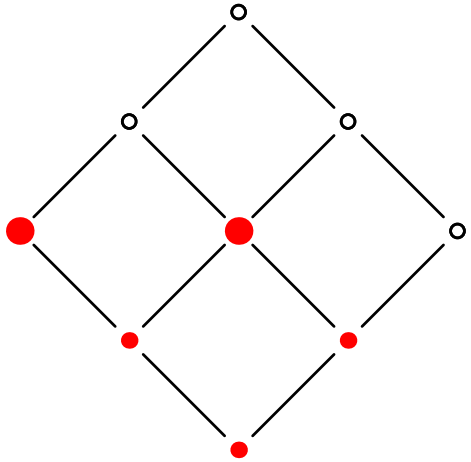


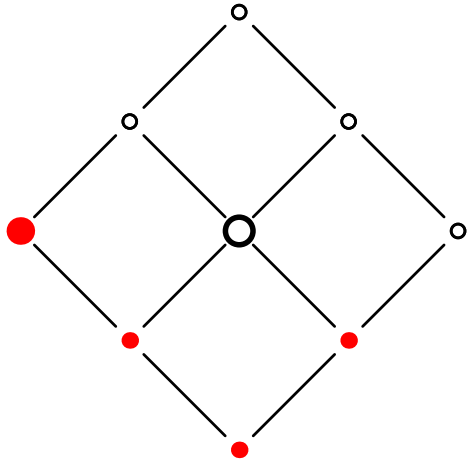


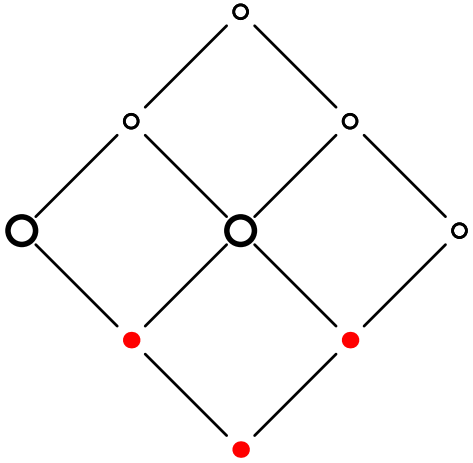


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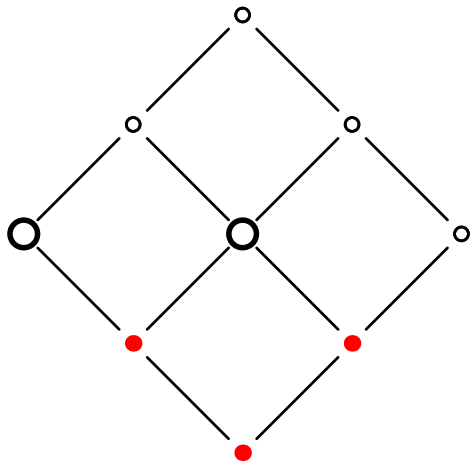








# Examples of Ungarian Markov Chains



**Remark**

*This Markov process corresponds to the well-known Multicorner Growth Process.*

# Central Question

**Definition**

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**Question**

*Given any lattice $L$, what is $\mathcal{E}(L)$?*

# $S_n$ as a Lattice

**Definition**

Given $\sigma \in S_n$,

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For example,

$$2\underline{41}7563 \gtrdot 2\underline{14}7563.$$

# $\mathrm{Av}_n(312)$ as a Lattice

## Definition

Consider the subset $\mathrm{Av}_n(312) \subset S_n$ of 312-avoiding permutations; that is, $\sigma$ for which there exist no $1 \leq i_1 < i_2 < i_3 \leq n$ satisfying $\sigma(i_1) > \sigma(i_3) > \sigma(i_2)$. This forms a lattice isomorphic to the well-known *Tamari lattice* $\mathrm{Tam}_n$.

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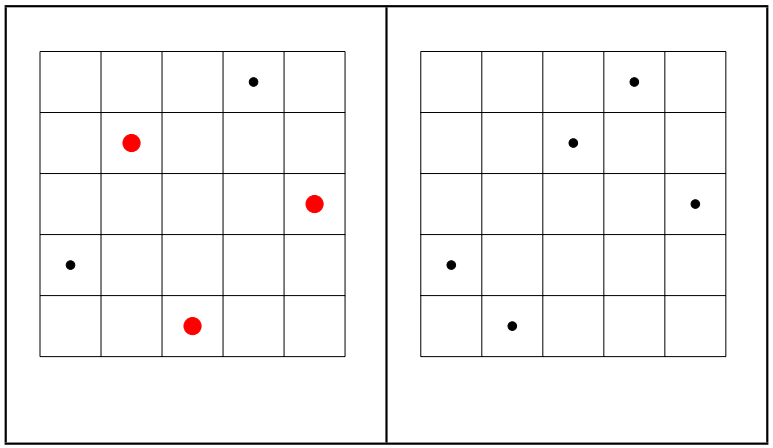


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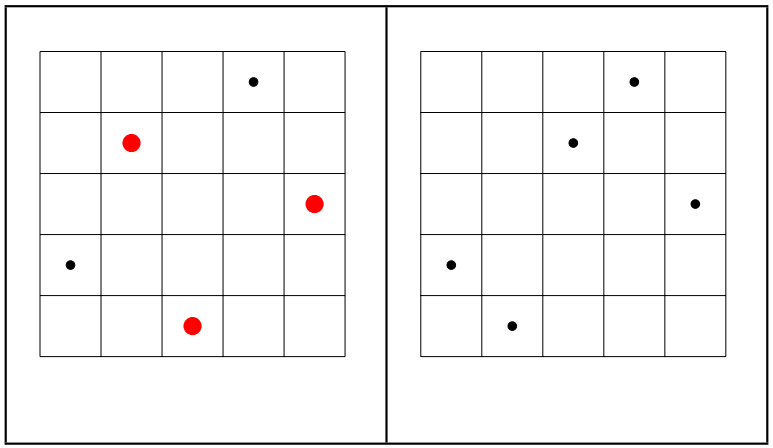


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## Remark

*For both* $S_n$ *and* $\mathrm{Av}_n(312)$, $\hat{1} = n(n-1)\cdots 1$ *and* $\hat{0} = 12\cdots n$.

# Ungarian Markov Chain on $S_n$

Fix $p \in (0, 1)$.

**Remark**

*Consider any $\sigma \in S_n$, and select each descent of $\sigma$ independently with probability $p$. A random Ungar move, applied to $\sigma$, reverses each the selected descending runs.*

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**Example**

Let $\sigma = 2415763$; then $\mathrm{Des}(\sigma) = \{2, 5, 6\}$. If $\{5, 6\}$ is selected (which occurs with probability $p^2(1-p)$), then applying the random Ungar move gives

$$2415\underline{763} \to 2415\underline{367}.$$

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Given $\sigma \in S_n$, define $\pi_\downarrow : S_n \to \mathrm{Av}_n(312)$ as follows. If there exist $1 \leq i < i+1 < j \leq n$ for which $\sigma(i) > \sigma(j) > \sigma(i+1)$, swap $\sigma(i)$ and $\sigma(i+1)$. Repeat until no such indices.

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**Remark**

*The Ungarian Markov chain on* $\mathrm{Tam}_n \cong \mathrm{Av}_n(312)$ *is the Markov chain with*

- *state space* $\mathrm{Av}_n(312)$, *and*
- *transitions given by the composition of a random Ungar move on* $S_n$ *and* $\pi_\downarrow$.

# Simulations

# Past Results

**Theorem (Defant–Li '23)**

$$n - 1 + o(1) \leq \mathcal{E}(S_n) \leq \frac{8}{p} n \log n + O(n)$$

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**Conjecture (Defant–Li '23)**

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$$\mathcal{E}(\mathrm{Tam}_n) = \frac{n}{e^{O(\log\log n)^3}}$$

# Acknowledgements

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# References


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